

23CSE214 – Operating Systems 2025-26 Even Semester

Case Study - eXpOS

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Stage 1 : Set Up & Installation

```
arunpravinap_24710@ubuntu:~$ sudo apt-get install libreadline-dev flex bison make gcc wget curl
[sudo] password for arunpravinap_24710:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
libreadline-dev is already the newest version (8.2-4build1).
flex is already the newest version (2.6.4-8.2build1).
bison is already the newest version (2:3.8.2+dfsg-1build2).
make is already the newest version (4.3-4.1build2).
gcc is already the newest version (4:13.2.0-7ubuntu1).
wget is already the newest version (1.21.4-1ubuntu4.1).
curl is already the newest version (8.5.0-2ubuntu10.6).
0 upgraded, 0 newly installed, 0 to remove and 269 not upgraded.
arunpravinap_24710@ubuntu:~$ curl -sSf https://raw.githubusercontent.com/eXpOSNite/expos-bootstrap/main/download.sh | sh
Downloading expl ..
expl downloaded.
Downloading spl ..
spl downloaded.
Downloading xfs-interface ..
xfs-interface downloaded.
Downloading xsm ..
xsm downloaded.
Download complete.
arunpravinap_24710@ubuntu:~$ cd $HOME/myexpos
arunpravinap_24710@ubuntu:~/myexpos$ make
make -C expl
make[1]: Entering directory '/home/arunpravinap_24710/myexpos/expl'
lex abstree.l
yacc -d abstree.y
```

Stage 2 : Understanding the Filesystem

```
arunpravinap_24710@ubuntu:~/myexpos$ cd xfs-interface
arunpravinap_24710@ubuntu:~/myexpos/xfs-interface$ ./xfs-interface
Unix-XFS Interface Version 2.0.
Type "help" for getting a list of commands.
# fdisk
Formatting Complete. "disk.xfs" created.
# exit
arunpravinap_24710@ubuntu:~/myexpos/xfs-interface$ df
Filesystem      1K-blocks    Used Available Use% Mounted on
tmpfs           1567480      2752    1564728   1% /run
efivarfs         128         45         79   37% /sys/firmware/efi/efivars
/dev/sda1       17718208 6207872 11510336  36% /cdrom
/cow            12450992 2071060  9750792  18% /
tmpfs           7837384         8    7837376   1% /dev/shm
tmpfs           5120         16     5104    1% /run/lock
tmpfs           7837384         4    7837380   1% /tmp
tmpfs           1567476      132    1567344   1% /run/user/1000
tmpfs           1567476      152    1567324   1% /run/user/1002
/dev/sda2       12450992 2071060  9750792  18% /media/arunpravinap_24710/casper-rw
arunpravinap_24710@ubuntu:~/myexpos/xfs-interface$ disk.xfs
disk.xfs: command not found
arunpravinap_24710@ubuntu:~/myexpos/xfs-interface$ ls
AUTHORS  disk.c  diskUtility.c  disk.xfs  exception.h  inode.c  interface.c  labels.c  LICENSE  memOrg.h  virtualDisk.c  xfs-interface
constants.h  disk.h  diskUtility.h  exception.c  fileSystem.h  inode.h  interface.h  labels.h  Makefile  README.md  virtualDisk.h
arunpravinap_24710@ubuntu:~/myexpos/xfs-interface$ ./xfs-interface
Unix-XFS Interface Version 2.0.
Type "help" for getting a list of commands.
# load --data $HOME/myexpos/sample.dat
# copy 3 4 $HOME/myexpos/inode_table.txt
# exit
arunpravinap_24710@ubuntu:~/myexpos/xfs-interface$ ./xfs-interface
Unix-XFS Interface Version 2.0.
Type "help" for getting a list of commands.
# copy 69 69 $HOME/myexpos/data.txt
#
```

sample.dat :

```
sample.dat
Amrita Vishwa Vidyapeetham, Coimbatore, is the esteemed headquarters of one of India's leading private research universities, renowned for blending modern academic excellence with traditional values. Located at the serene foothills of the Western Ghats in Ettimadai, the 400-acre campus provides a world-class environment for over 12,000 students. As a NAAC A++ accredited institution, it consistently ranks among the top universities in India according to NIRF. The campus is a multidisciplinary powerhouse, hosting premier schools for Engineering, Business, and Mass Communication, all while emphasizing "Education for Life" through its focus on research, compassion-driven innovation, and global sustainability.
```

data.txt :

```
Amrita Vishwa V
idyapeetham, Co
imbatore, is th
e esteemed head
quarters of one
of India's l
eading private
research univer
sities,
renowned for bl
ending modern a
cademic excelle
nce with tradit
ional values. L
located at the
```

Stage 1 - Assignment 1 : Copy the contents of Root File (from Block 5 of XFS disk) to a UNIX file `$HOME/myexpos/root_file.txt` and verify that an entry for *sample.dat* is made in it also.

```
arunpravinap_24710@ubuntu:~/myexpos/xfs-interface$ ./xfs-interface
Unix-XFS Interace Version 2.0.
Type "help" for getting a list of commands.
# copy 5 5 $HOME/myexpos/root_file.txt
# exit

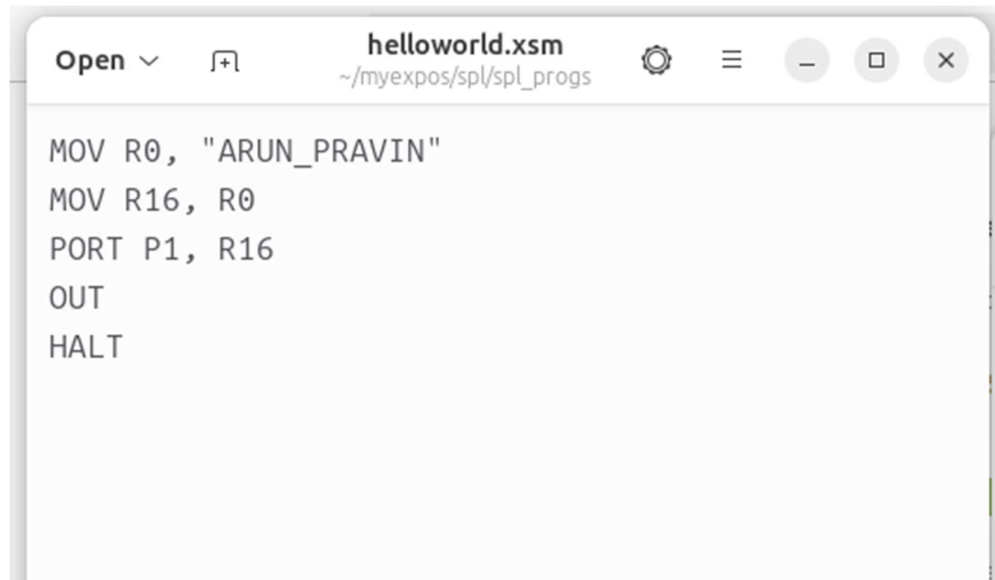
arunpravinap_24710@ubuntu:~/myexpos/xfs-interface$ cat $HOME/myexpos/root_file.txt
root
512
1
-1
-1
-1
-1
-1
sample.dat
51
2
-1
-1
-1
-1
-1
-1
0
```

Stage 1 - Assignment 2 : Delete the *sample.dat* from the XSM machine using xfs-interface and note the changes for the entries for this file in *inode table*, *root file* and *disk free list* .

```
arunpravinap_24710@ubuntu:~/myexpos/xfs-interface$ ./xfs-interface
Unix-XFS Interace Version 2.0.
Type "help" for getting a list of commands.
# rm sample.dat
```

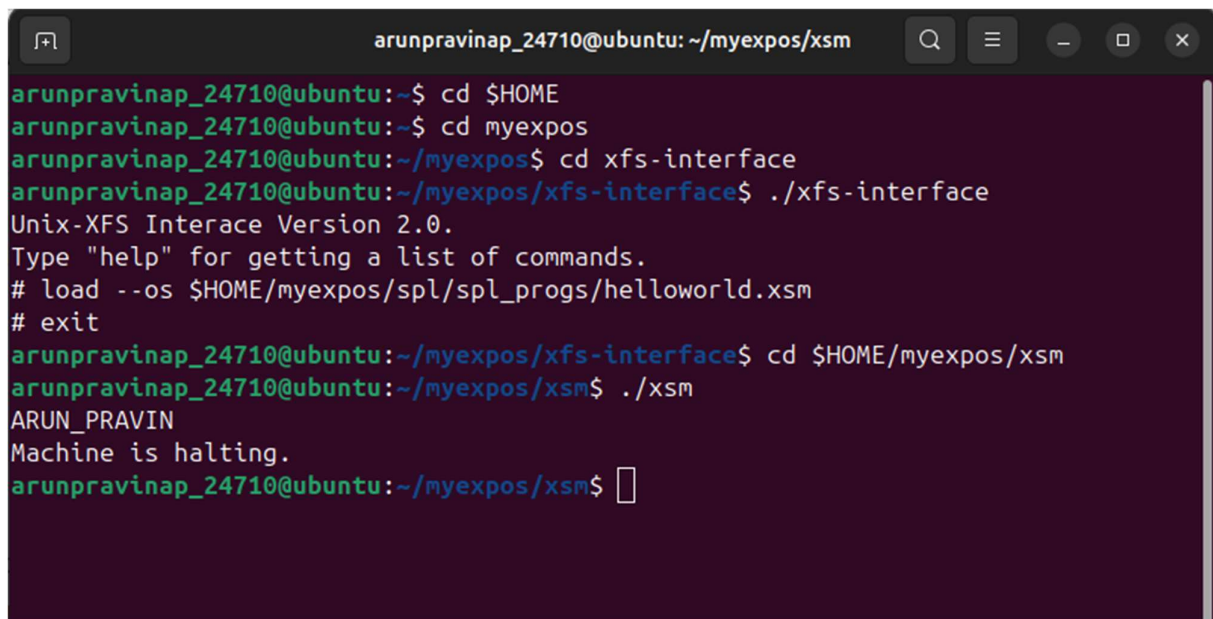
64	-	1
65	-	1
66	-	1
67	-	1
68	-	1
69	-	0
70	-	0
71	-	0
72	-	0
73	-	0
74	-	0
75	-	0
76	-	0
77	-	0
78	-	0
79	-	0
80	-	0
81	-	0
82	-	0
83	-	0
84	-	0
85	-	0
86	-	0
87	-	0
88	-	0
89	-	0
90	-	0
91	-	0
92	-	0
93	-	0
94	-	0
95	-	0
96	-	0
97	-	0

Step 3 : BootStrap Loader



A screenshot of a text editor window titled 'helloworld.xsm' with the path '~/.myexpos/spl/spl_progs'. The window contains the following assembly code:

```
MOV R0, "ARUN_PRAVIN"  
MOV R16, R0  
PORT P1, R16  
OUT  
HALT
```



A screenshot of a terminal window showing the execution of the 'helloworld.xsm' program. The user is 'arunpravinap_24710' on an 'ubuntu' machine. The terminal shows the following sequence of commands and output:

```
arunpravinap_24710@ubuntu:~$ cd $HOME  
arunpravinap_24710@ubuntu:~$ cd myexpos  
arunpravinap_24710@ubuntu:~/myexpos$ cd xfs-interface  
arunpravinap_24710@ubuntu:~/myexpos/xfs-interface$ ./xfs-interface  
Unix-XFS Interace Version 2.0.  
Type "help" for getting a list of commands.  
# load --os $HOME/myexpos/spl/spl_progs/helloworld.xsm  
# exit  
arunpravinap_24710@ubuntu:~/myexpos/xfs-interface$ cd $HOME/myexpos/xsm  
arunpravinap_24710@ubuntu:~/myexpos/xsm$ ./xsm  
ARUN_PRAVIN  
Machine is halting.  
arunpravinap_24710@ubuntu:~/myexpos/xsm$
```

Stage 3 - Assignment 1 : Write an assembly program to print numbers from 1 to 20 and run it as the OS Startup code.

```
stage3_assignment1.xsm
MOV R0, 1      ; counter = 1
MOV R1, 21     ; limit = 21

LOOP:
MOV R16, R0
PORT P1, R16
OUT

ADD R0, 1      ; R0 = R0 + 1

MOV R2, R0     ; copy counter
LT R2, R1      ; R2 = 1 if R0 < 21 else 0
JNZ R2, LOOP   ; if true, repeat

HALT
```

```
arunpravinap_24710@ubuntu:~/myexpos/xsm$ cd $HOME/myexpos/xfs-interface
arunpravinap_24710@ubuntu:~/myexpos/xfs-interface$ ./xfs-interface
Unix-XFS Interface Version 2.0.
Type "help" for getting a list of commands.
# load --os $HOME/myexpos/spl/spl_progs/stage3_assignment1.xsm
# exit
arunpravinap_24710@ubuntu:~/myexpos/xfs-interface$ cd $HOME/myexpos/xsm
arunpravinap_24710@ubuntu:~/myexpos/xsm$ ./xsm
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
Machine is halting.
arunpravinap_24710@ubuntu:~/myexpos/xsm$
```

Stage 4 : Learning the SPL Language

```
arunpravinap_24710@ubuntu: ~/myexpos/xsm
arunpravinap_24710@ubuntu:~/myexpos/xsm$ cd $HOME/myexpos/spl
arunpravinap_24710@ubuntu:~/myexpos/spl$ ./spl $HOME/myexpos/spl/spl_progs/oddnos.spl
arunpravinap_24710@ubuntu:~/myexpos/spl$ cd $HOME/myexpos/xfs-interface
arunpravinap_24710@ubuntu:~/myexpos/xfs-interface$ ./xfs-interface
Unix-XFS Interace Version 2.0.
Type "help" for getting a list of commands.
# load --os $HOME/myexpos/spl/spl_progs/oddnos.xsm
# exit
arunpravinap_24710@ubuntu:~/myexpos/xfs-interface$ cd $HOME/myexpos/xsm
arunpravinap_24710@ubuntu:~/myexpos/xsm$ ./xsm
1
3
5
7
9
11
13
15
17
19
Machine is halting.
arunpravinap_24710@ubuntu:~/myexpos/xsm$
```

oddnos.spl – SPL Code:

```
alias counter R0;
counter = 0;
while(counter <= 20) do
  if(counter%2 != 0) then
    print counter;
  endif;
  counter = counter + 1;
endwhile;
```

oddnos.xsm – Compiled XSM Code:

```
MOV R0, 0
_L1:
MOV R16, 20
GE R16, R0
JZ R16, _L2
MOV R16, R0
MOD R16, 2
MOV R17, 0
NE R16, R17
JZ R16, _L3
MOV R16, R0
PORT P1, R16
OUT
JMP _L4
_L3:
_L4:
MOV R16, R0
ADD R16, 1
MOV R0, R16
JMP _L1
_L2:
HALT
```


Stage 4 - Assignment 1 : Write the spl program to print sum of squares of the first 20 natural numbers. Load it using xfs interface and run the in the machine.

```
arunpravinap_24710@ubuntu:~/myexpos/xsm$ cd $HOME/myexpos/spl
arunpravinap_24710@ubuntu:~/myexpos/spl$ ./spl $HOME/myexpos/spl/spl_progs/stage4_assignment1.spl
arunpravinap_24710@ubuntu:~/myexpos/spl$ cd $HOME/myexpos/xfs-interface
arunpravinap_24710@ubuntu:~/myexpos/xfs-interface$ ./xfs-interface
Unix-XFS Interface Version 2.0.
Type "help" for getting a list of commands.
# load --os $HOME/myexpos/spl/spl_progs/stage4_assignment1.xsm
# exit
arunpravinap_24710@ubuntu:~/myexpos/xfs-interface$ cd $HOME/myexpos/xsm
arunpravinap_24710@ubuntu:~/myexpos/xsm$ ./xsm
2870
Machine is halting.
arunpravinap_24710@ubuntu:~/myexpos/xsm$
```

stage4_assignment1.spl SPL code:

```
alias i R0;
alias sum R1;

i = 1;
sum = 0;

while(i <= 20) do
    sum = sum + (i * i);
    i = i + 1;
endwhile;

print sum;
```

Compiled xsm Code:

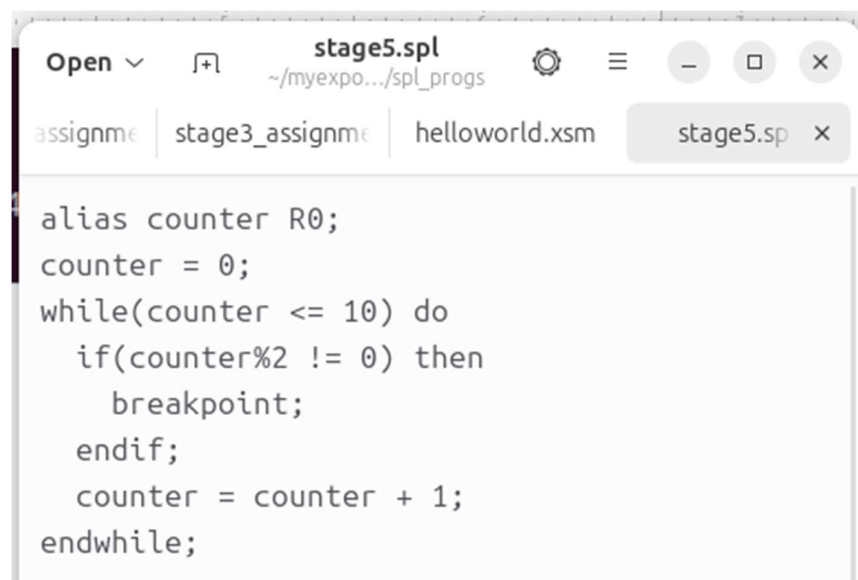
```
Open ▾ stage4_assign...
~/myexpo.../spl_progs

stage4_assignment1 x stage3_assignment1 helloworld.xsm

MOV R0, 1
MOV R1, 0
_L1:
MOV R16, 20
GE R16, R0
JZ R16, _L2
MOV R16, R0
MUL R16, R0
ADD R16, R1
MOV R1, R16
MOV R16, R0
ADD R16, 1
MOV R0, R16
JMP _L1
_L2:
MOV R16, R1
PORT P1, R16
OUT
HALT
```

Stage 5 : XSM Debugging

```
arunpravinap_24710@ubuntu:~/myexpos/xsm$ cd $HOME/myexpos/spl
arunpravinap_24710@ubuntu:~/myexpos/spl$ ./spl $HOME/myexpos/spl/spl_progs/stage5.spl
arunpravinap_24710@ubuntu:~/myexpos/spl$ cd $HOME/myexpos/xfs-interface
arunpravinap_24710@ubuntu:~/myexpos/xfs-interface$ ./xfs-interface
Unix-XFS Interace Version 2.0.
Type "help" for getting a list of commands.
# load --os $HOME/myexpos/spl/spl_progs/stage5.xsm
# exit
arunpravinap_24710@ubuntu:~/myexpos/xfs-interface$ cd $HOME/myexpos/xsm
arunpravinap_24710@ubuntu:~/myexpos/xsm$ ./xsm --debug
Previous instruction at IP = 530: BRKP
Mode: KERNEL      PID: -1
Next instruction at IP = 532, Page No. = 1: JMP 534
debug> reg
R0: 1   R1:    R2:    R3:    R4:
R5:    R6:    R7:    R8:    R9:
R10:   R11:   R12:   R13:   R14:
R15:   R16: 1  R17: 0  R18:   R19:
P0:    P1:    P2:    P3:
BP:    SP:    IP: 532 PTBR:   PTLR:
EIP:   EC:    EPN:   EMA:
debug> mem 1
Page: 1
Written to file mem
debug> s
Previous instruction at IP = 532: JMP 534
Mode: KERNEL      PID: -1
Next instruction at IP = 534, Page No. = 1: MOV R16,R0
debug> c
Previous instruction at IP = 530: BRKP
Mode: KERNEL      PID: -1
Next instruction at IP = 532, Page No. = 1: JMP 534
debug> 
```



```
alias counter R0;
counter = 0;
while(counter <= 10) do
    if(counter%2 != 0) then
        breakpoint;
    endif;
    counter = counter + 1;
endwhile;
```


Stage 6 – Running a User Program

- Loading *squares.xsm* as the init program

```
arunpravinap_24710@ubuntu:~/myexpos/xf-interface$ nano $HOME/myexpos/expl/expl_progs/squares.xsm
arunpravinap_24710@ubuntu:~/myexpos/xf-interface$ ./xf-interface
Unix-XFS Interface Version 2.0.
Type "help" for getting a list of commands.
# load --init ../expl/expl_progs/squares.xsm
# exit
```

- The halt program

```
arunpravinap_24710@ubuntu:~/myexpos/xf-interface$ nano $HOME/myexpos/spl/spl_progs/haltprog.spl
arunpravinap_24710@ubuntu:~/myexpos/xf-interface$ cd $HOME/myexpos/spl
arunpravinap_24710@ubuntu:~/myexpos/spl$ ./spl spl_progs/haltprog.spl
arunpravinap_24710@ubuntu:~/myexpos/spl$ ls spl_progs | grep haltprog
haltprog.spl
haltprog.xsm
arunpravinap_24710@ubuntu:~/myexpos/spl$ cd ../xf-interface
arunpravinap_24710@ubuntu:~/myexpos/xf-interface$ ./xf-interface
Unix-XFS Interface Version 2.0.
Type "help" for getting a list of commands.
# load --int=10 ../spl/spl_progs/haltprog.xsm
# exit
```

- Loading the OS startup Program

```
arunpravinap_24710@ubuntu:~/myexpos/xf-interface$ nano $HOME/myexpos/spl/spl_progs/os_startup.spl
arunpravinap_24710@ubuntu:~/myexpos/xf-interface$ cd $HOME/myexpos/spl
arunpravinap_24710@ubuntu:~/myexpos/spl$ ./spl spl_progs/os_startup.spl
arunpravinap_24710@ubuntu:~/myexpos/spl$ ls spl_progs | grep os_startup
os_startup.spl
os_startup.xsm
arunpravinap_24710@ubuntu:~/myexpos/spl$ cd ../xf-interface
arunpravinap_24710@ubuntu:~/myexpos/xf-interface$ ./xf-interface
Unix-XFS Interface Version 2.0.
Type "help" for getting a list of commands.
# load --os ../spl/spl_progs/os_startup.xsm
# exit
```

- O/P : Now we can see the contents of R1 in each stage

```
Previous instruction at IP = 12: BRKP
Mode: USER      PID: 0
Next instruction at IP = 14, Page No. = 0: ADD R0,1
debug> s
Previous instruction at IP = 14: ADD R0,1
Mode: USER      PID: 0
Next instruction at IP = 16, Page No. = 0: JMP 2
debug> reg
R0: 2   R1: 1   R2: 1   R3:    R4:
R5:    R6:    R7:    R8:    R9:
R10:   R11:   R12:   R13:   R14:
R15:   R16: 1024 R17:   R18:   R19:
P0:    P1:    P2:    P3:
BP:    SP: 1023 IP: 16 PTBR: 29696 PTLR: 3
EIP:   EC:    EPN:   EMA:
debug> c
Previous instruction at IP = 12: BRKP
Mode: USER      PID: 0
Next instruction at IP = 14, Page No. = 0: ADD R0,1
debug> s
Previous instruction at IP = 14: ADD R0,1
Mode: USER      PID: 0
Next instruction at IP = 16, Page No. = 0: JMP 2
debug> []
```

Assignment 1 : Change virtual memory model such that code occupies logical pages 4 and 5 and the stack lies in logical page 8. You will have to modify the user program as well as the os startup code.

- User Program and Startup code

```
// Squares of first 5 numbers
// Code starts at logical page 4 → address 2048

2048 MOV R0, 1      // n = 1

2050 MOV R2, 5      // limit = 5
2052 GE R2, R0      // if n > 5 ?
2054 JZ R2, 2066    // jump to exit if true

2056 MOV R1, R0     // R1 = n
2058 MUL R1, R0     // R1 = n * n
2060 BRKP          // breakpoint to check R1

2062 ADD R0, 1      // n = n + 1
2064 JMP 2050       // repeat loop

2066 INT 10        // exit system call
```

```
MOV R0, 1

MOV R2, 5
GE R2, R0
JZ R2, 2070

MOV R1, R0
MUL R1, R0
BRKP

ADD R0, 1
JMP 2050

INT 10
```

```
loadi(65,7);
loadi(66,8);

loadi(22,35);
loadi(23,36);

loadi(2,15);
loadi(3,16);

PTBR = PAGE_TABLE_BASE;
PTLR = 9;

// Code pages: logical 4 and 5
[PTBR + 8] = 65;
[PTBR + 9] = "0100";

[PTBR + 10] = 66;
[PTBR + 11] = "0100";

// Stack page: logical 8
[PTBR + 16] = 76;
[PTBR + 17] = "0110";

// Set stack pointer
SP = 8 * 512;

// Push starting IP (2048) to user stack
[76 * 512] = 2048;

ireturn;
```

```

arunpravinap_24710@ubuntu-Exp:~/myexpos/xsm$ vim $HOME/myexpos/expl/expl_progs/squares.xsm
arunpravinap_24710@ubuntu-Exp:~/myexpos/xsm$ cd $HOME/myexpos/xfs-interface
arunpravinap_24710@ubuntu-Exp:~/myexpos/xfs-interface$ ./xfs-interface
Unix-XFS Interface Version 2.0.
Type "help" for getting a list of commands.
# load --init $HOME/myexpos/expl/expl_progs/squares.xsm
# exit
arunpravinap_24710@ubuntu-Exp:~/myexpos/xfs-interface$ vim $HOME/myexpos/spl/spl_progs/haltprog.spl
arunpravinap_24710@ubuntu-Exp:~/myexpos/xfs-interface$ cd ../spl
arunpravinap_24710@ubuntu-Exp:~/myexpos/spl$ ./spl spl_progs/haltprog.spl
arunpravinap_24710@ubuntu-Exp:~/myexpos/spl$ cd ../xfs-interface
arunpravinap_24710@ubuntu-Exp:~/myexpos/xfs-interface$ ./xfs-interface
Unix-XFS Interface Version 2.0.
Type "help" for getting a list of commands.
# load --init=10 ../spl/spl_progs/haltprog.xsm
# exit
arunpravinap_24710@ubuntu-Exp:~/myexpos/xfs-interface$ ./xfs-interface
Unix-XFS Interface Version 2.0.
Type "help" for getting a list of commands.
# load --exhandler ../spl/spl_progs/haltprog.xsm
# exit
arunpravinap_24710@ubuntu-Exp:~/myexpos/xfs-interface$ vim $HOME/myexpos/spl/spl_progs/os_startup.spl
arunpravinap_24710@ubuntu-Exp:~/myexpos/xfs-interface$ cd ../spl
arunpravinap_24710@ubuntu-Exp:~/myexpos/spl$ ./spl spl_progs/os_startup.spl
arunpravinap_24710@ubuntu-Exp:~/myexpos/spl$ cd ../xfs-interface
arunpravinap_24710@ubuntu-Exp:~/myexpos/xfs-interface$ ./xfs-interface
Unix-XFS Interface Version 2.0.
Type "help" for getting a list of commands.
# load --os ../spl/spl_progs/os_startup.xsm
# exit
arunpravinap_24710@ubuntu-Exp:~/myexpos/xsm$ cd ../xsm
arunpravinap_24710@ubuntu-Exp:~/myexpos/xsm$ ./xsm --debug -timer 0

```

```

Previous instruction at IP = 2060: BRKP
Mode: USER PID: 0
Next instruction at IP = 2062, Page No. = 4: ADD R0,1
debug> reg
R0: 1 R1: 1 R2: 1 R3: R4:
R5: R6: R7: R8: R9:
R10: R11: R12: R13: R14:
R15: R16: 38912 R17: R18: R19:
P0: P1: P2: P3:
BP: SP: 4095 IP: 2062 PTBR: 29696 PTLR: 9
EIP: EC: EPN: EMA:
debug> c
Previous instruction at IP = 2060: BRKP
Mode: USER PID: 0
Next instruction at IP = 2062, Page No. = 4: ADD R0,1
debug> reg
R0: 2 R1: 4 R2: 1 R3: R4:
R5: R6: R7: R8: R9:
R10: R11: R12: R13: R14:
R15: R16: 38912 R17: R18: R19:
P0: P1: P2: P3:
BP: SP: 4095 IP: 2062 PTBR: 29696 PTLR: 9
EIP: EC: EPN: EMA:
debug> reg
R0: 2 R1: 4 R2: 1 R3: R4:
R5: R6: R7: R8: R9:
R10: R11: R12: R13: R14:
R15: R16: 38912 R17: R18: R19:
P0: P1: P2: P3:
BP: SP: 4095 IP: 2062 PTBR: 29696 PTLR: 9
EIP: EC: EPN: EMA:
debug> c
Previous instruction at IP = 2060: BRKP
Mode: USER PID: 0
Next instruction at IP = 2062, Page No. = 4: ADD R0,1
debug> reg
R0: 3 R1: 9 R2: 1 R3: R4:
R5: R6: R7: R8: R9:
R10: R11: R12: R13: R14:
R15: R16: 38912 R17: R18: R19:
P0: P1: P2: P3:
BP: SP: 4095 IP: 2062 PTBR: 29696 PTLR: 9
EIP: EC: EPN: EMA:
debug> c
Previous instruction at IP = 2060: BRKP
Mode: USER PID: 0
Next instruction at IP = 2062, Page No. = 4: ADD R0,1

```

```

debug> reg
R0: 4   R1: 16  R2: 1   R3:      R4:
R5:      R6:      R7:      R8:      R9:
R10:     R11:     R12:     R13:     R14:
R15:     R16: 38912   R17:     R18:     R19:
P0:      P1:      P2:      P3:
BP:      SP: 4095    IP: 2062      PTBR: 29696    PTLR: 9
EIP:     EC:      EPN:     EMA:
debug> c
Previous instruction at IP = 2060: BRKP
Mode: USER      PID: 0
Next instruction at IP = 2062, Page No. = 4: ADD R0,1
debug> reg
R0: 5   R1: 25  R2: 1   R3:      R4:
R5:      R6:      R7:      R8:      R9:
R10:     R11:     R12:     R13:     R14:
R15:     R16: 38912   R17:     R18:     R19:
P0:      P1:      P2:      P3:
BP:      SP: 4095    IP: 2062      PTBR: 29696    PTLR: 9
EIP:     EC:      EPN:     EMA:
debug> c
Machine is halting.

```

Stage 7 : XEXE Executable File Format and ABI

squares.xsm

```

0
2056
0
0
0
0
0
0
0
0
MOV R0, 1
MOV R2, 5
GE R2, R0
JZ R2, 2074
MOV R1, R0
MUL R1, R0
BRKP
ADD R0, 1
JMP 2058
INT 10

```

os_startup.spl

```

loadi(63,13);
loadi(64,14);

loadi(65,7);
loadi(66,8);

loadi(22,35);
loadi(23,36);

loadi(2,15);
loadi(3,16);

PTBR = PAGE_TABLE_BASE;
PTLR = 10;

[PTBR+0] = 63; [PTBR+1] = "0100";
[PTBR+2] = 64; [PTBR+3] = "0100";

[PTBR+4] = 78; [PTBR+5] = "0110";
[PTBR+6] = 79; [PTBR+7] = "0110";

[PTBR+8] = 65; [PTBR+9] = "0100";
[PTBR+10] = 66; [PTBR+11] = "0100";
[PTBR+12] = -1; [PTBR+13] = "0000";
[PTBR+14] = -1; [PTBR+15] = "0000";

[PTBR+16] = 76; [PTBR+17] = "0110";
[PTBR+18] = 77; [PTBR+19] = "0110";

SP = 9 * 512 + 511;
[77*512+511] = [65*512 + 1];

ireturn;

```



```

arunpravinap_24710@ubuntu-Exp:~/myexpos/xsm$ cd ../xfs-interface
arunpravinap_24710@ubuntu-Exp:~/myexpos/xfs-interface$ ./xfs-interface
Unix-XFS Interface Version 2.0.
Type "help" for getting a list of commands.
# load --library ../expl/library.lib
# exit
arunpravinap_24710@ubuntu-Exp:~/myexpos/xsm$ vim $HOME/myexpos/expl/expl_progs/squares.xsm
arunpravinap_24710@ubuntu-Exp:~/myexpos/xfs-interface$
Unix-XFS Interface Version 2.0.
Type "help" for getting a list of commands.
# load --init ../expl/expl_progs/squares.xsm
# exit
arunpravinap_24710@ubuntu-Exp:~/myexpos/xfs-interface$ vim $HOME/myexpos/spl/spl_progs/os_startup.sp
arunpravinap_24710@ubuntu-Exp:~/myexpos/xfs-interface$ cd ../spl
arunpravinap_24710@ubuntu-Exp:~/myexpos/spl$ ./spl spl_progs/os_startup.spl
arunpravinap_24710@ubuntu-Exp:~/myexpos/spl$ cd ../xfs-interface
arunpravinap_24710@ubuntu-Exp:~/myexpos/xfs-interface$ ./xfs-interface
Unix-XFS Interface Version 2.0.
Type "help" for getting a list of commands.
# load --os ../spl/spl_progs/os_startup.xsm
# exit
arunpravinap_24710@ubuntu-Exp:~/myexpos/xsm$ cd ../xsm
arunpravinap_24710@ubuntu-Exp:~/myexpos/xsm$ ./xsm --debug -timer 0
Previous instruction at IP = 2070: BRKP
Mode: USER      PID: 0
Next instruction at IP = 2072, Page No. = 4: ADD R0,1
arunpravinap_24710@ubuntu-Exp:~/myexpos/xsm$ cd ../xsm
arunpravinap_24710@ubuntu-Exp:~/myexpos/xsm$ ./xsm --debug--timer 0

```

Output:

```

Previous instruction at IP = 2068: BRKP
Mode: USER      PID: 0
Next instruction at IP = 2070, Page No. = 4: ADD R0,1
debug> reg
R0: 1  R1: 1  R2: 1  R3:  R4:
R5:  R6:  R7:  R8:  R9:
R10:  R11:  R12:  R13:  R14:
R15:  R16: 39935  R17: 2056  R18:  R19:
P0:  P1:  P2:  P3:
BP:  SP: 5118  IP: 2070  PTBR: 29696  PTLR: 10
EIP:  EC:  EPN:  EMA:
debug> c
Previous instruction at IP = 2068: BRKP
Mode: USER      PID: 0
Next instruction at IP = 2070, Page No. = 4: ADD R0,1
debug> reg
R0: 2  R1: 4  R2: 1  R3:  R4:
R5:  R6:  R7:  R8:  R9:
R10:  R11:  R12:  R13:  R14:
R15:  R16: 39935  R17: 2056  R18:  R19:
P0:  P1:  P2:  P3:
BP:  SP: 5118  IP: 2070  PTBR: 29696  PTLR: 10
EIP:  EC:  EPN:  EMA:
debug> c
Previous instruction at IP = 2068: BRKP
Mode: USER      PID: 0
Next instruction at IP = 2070, Page No. = 4: ADD R0,1
debug> reg
R0: 3  R1: 9  R2: 1  R3:  R4:
R5:  R6:  R7:  R8:  R9:
R10:  R11:  R12:  R13:  R14:
R15:  R16: 39935  R17: 2056  R18:  R19:
P0:  P1:  P2:  P3:
BP:  SP: 5118  IP: 2070  PTBR: 29696  PTLR: 10
EIP:  EC:  EPN:  EMA:

```

```

debug> c
Previous instruction at IP = 2068: BRKP
Mode: USER      PID: 0
Next instruction at IP = 2070, Page No. = 4: ADD R0,1
debug> reg
R0: 4  R1: 16  R2: 1  R3:  R4:
R5:  R6:  R7:  R8:  R9:
R10:  R11:  R12:  R13:  R14:
R15:  R16: 39935  R17: 2056  R18:  R19:
P0:  P1:  P2:  P3:
BP:  SP: 5118  IP: 2070  PTBR: 29696  PTLR: 10
EIP:  EC:  EPN:  EMA:
debug> c
Previous instruction at IP = 2068: BRKP
Mode: USER      PID: 0
Next instruction at IP = 2070, Page No. = 4: ADD R0,1
debug> reg
R0: 5  R1: 25  R2: 1  R3:  R4:
R5:  R6:  R7:  R8:  R9:
R10:  R11:  R12:  R13:  R14:
R15:  R16: 39935  R17: 2056  R18:  R19:
P0:  P1:  P2:  P3:
BP:  SP: 5118  IP: 2070  PTBR: 29696  PTLR: 10
EIP:  EC:  EPN:  EMA:
debug> c
Machine is halting.

```

Stage 7 Assignment 1 : Change the user program to compute cubes of the first five numbers.

Cubes.xsm

```
0
2056
0
0
0
0
0
0
0

MOV R0, 1

MOV R2, 5
GE R2, R0
JZ R2, 2078

MOV R1, R0
MUL R1, R0
MUL R1, R0
BRKP

ADD R0, 1
JMP 2058

INT 10
```

os_startup.spl

```
loadi(63,13);
loadi(64,14);

loadi(65,7);
loadi(66,8);

loadi(22,35);
loadi(23,36);

loadi(2,15);
loadi(3,16);

PTBR = PAGE_TABLE_BASE;
PTLR = 10;

[PTBR+0] = 63; [PTBR+1] = "0100";
[PTBR+2] = 64; [PTBR+3] = "0100";

[PTBR+4] = 78; [PTBR+5] = "0110";
[PTBR+6] = 79; [PTBR+7] = "0110";

[PTBR+8] = 65; [PTBR+9] = "0100";
[PTBR+10] = 66; [PTBR+11] = "0100";
[PTBR+12] = -1; [PTBR+13] = "0000";
[PTBR+14] = -1; [PTBR+15] = "0000";

[PTBR+16] = 76; [PTBR+17] = "0110";
[PTBR+18] = 77; [PTBR+19] = "0110";

SP = 8 * 512;
[76*512] = [65*512 + 1];

ireturn;
```

```
arunpravinap_24710@ubuntu-Exp:~/myexpos/xsm$ cd ../xfs-interface
arunpravinap_24710@ubuntu-Exp:~/myexpos/xfs-interface$ ./xfs-interface
Unix-XFS Interface Version 2.0.
Type "help" for getting a list of commands.
# load --library ../expl/library.lib
# exit
arunpravinap_24710@ubuntu-Exp:~/myexpos/xsm$ vim $HOME/myexpos/expl/expl_progs/cubes.xsm
arunpravinap_24710@ubuntu-Exp:~/myexpos/xfs-interface$ ./xfs-interface
Unix-XFS Interface Version 2.0.
Type "help" for getting a list of commands.
# load --init ../expl/expl_progs/cubes.xsm
# exit
arunpravinap_24710@ubuntu-Exp:~/myexpos/xsm$ vim $HOME/myexpos/spl/spl_progs/os_startup.spl
arunpravinap_24710@ubuntu-Exp:~/myexpos/spl$ cd ../spl
arunpravinap_24710@ubuntu-Exp:~/myexpos/spl$ ./spl spl_progs/os_startup.spl
arunpravinap_24710@ubuntu-Exp:~/myexpos/xfs-interface$ ./xfs-interface
Unix-XFS Interface Version 2.0.
Type "help" for getting a list of commands.
# load --os ../spl/spl_progs/os_startup.xsm
# exit
arunpravinap_24710@ubuntu-Exp:~/myexpos/xsm$ cd ../xsm
arunpravinap_24710@ubuntu-Exp:~/myexpos/xsm$ ./xsm --debug -timer 0
Previous instruction at IP = 2070: BRKP
Mode: USER PID: 0
Next instruction at IP = 2072, Page No. = 4: ADD R0,1
arunpravinap_24710@ubuntu-Exp:~/myexpos/xsm$ cd ../xsm
arunpravinap_24710@ubuntu-Exp:~/myexpos/xsm$ ./xsm --debug -timer 0
█
```


Output:

```
Previous instruction at IP = 2070: BRKP
Mode: USER      PID: 0
Next instruction at IP = 2072, Page No. = 4: ADD R0,1
debug> reg
R0: 1  R1: 1  R2: 1  R3:  R4:
R5:  R6:  R7:  R8:  R9:
R10:  R11:  R12:  R13:  R14:
R15:  R16: 39935  R17: 2056  R18:  R19:
P0:  P1:  P2:  P3:
BP:  SP: 5118  IP: 2072  PTBR: 29696  PTLR: 10
EIP:  EC:  EPN:  EMA:
debug> c
Previous instruction at IP = 2070: BRKP
Mode: USER      PID: 0
Next instruction at IP = 2072, Page No. = 4: ADD R0,1
debug> reg
R0: 2  R1: 8  R2: 1  R3:  R4:
R5:  R6:  R7:  R8:  R9:
R10:  R11:  R12:  R13:  R14:
R15:  R16: 39935  R17: 2056  R18:  R19:
P0:  P1:  P2:  P3:
BP:  SP: 5118  IP: 2072  PTBR: 29696  PTLR: 10
EIP:  EC:  EPN:  EMA:
debug> c
Previous instruction at IP = 2070: BRKP
Mode: USER      PID: 0
Next instruction at IP = 2072, Page No. = 4: ADD R0,1
debug> reg
R0: 3  R1: 27  R2: 1  R3:  R4:
R5:  R6:  R7:  R8:  R9:
R10:  R11:  R12:  R13:  R14:
R15:  R16: 39935  R17: 2056  R18:  R19:
P0:  P1:  P2:  P3:
BP:  SP: 5118  IP: 2072  PTBR: 29696  PTLR: 10
EIP:  EC:  EPN:  EMA:
```

```
debug> reg
R0: 3  R1: 27  R2: 1  R3:  R4:
R5:  R6:  R7:  R8:  R9:
R10:  R11:  R12:  R13:  R14:
R15:  R16: 39935  R17: 2056  R18:  R19:
P0:  P1:  P2:  P3:
BP:  SP: 5118  IP: 2072  PTBR: 29696  PTLR: 10
EIP:  EC:  EPN:  EMA:
debug> c
Previous instruction at IP = 2070: BRKP
Mode: USER      PID: 0
Next instruction at IP = 2072, Page No. = 4: ADD R0,1
debug> reg
R0: 4  R1: 64  R2: 1  R3:  R4:
R5:  R6:  R7:  R8:  R9:
R10:  R11:  R12:  R13:  R14:
R15:  R16: 39935  R17: 2056  R18:  R19:
P0:  P1:  P2:  P3:
BP:  SP: 5118  IP: 2072  PTBR: 29696  PTLR: 10
EIP:  EC:  EPN:  EMA:
debug> c
Previous instruction at IP = 2070: BRKP
Mode: USER      PID: 0
Next instruction at IP = 2072, Page No. = 4: ADD R0,1
debug> reg
R0: 5  R1: 125  R2: 1  R3:  R4:
R5:  R6:  R7:  R8:  R9:
R10:  R11:  R12:  R13:  R14:
R15:  R16: 39935  R17: 2056  R18:  R19:
P0:  P1:  P2:  P3:
BP:  SP: 5118  IP: 2072  PTBR: 29696  PTLR: 10
EIP:  EC:  EPN:  EMA:
debug> c
Machine is halting.
```

Stage 8 : TIMERS :

os_startup.xsm :

```
print "TIMER";
ireturn;
```

```
ubuntu@arunpravinap24710:~/myexpos/xsm$ cd $HOME/myexpos/spl
ubuntu@arunpravinap24710:~/myexpos/spl$ ./spl $HOME/myexpos/spl/spl_progs/sample_timer.spl
ubuntu@arunpravinap24710:~/myexpos/spl$ cd $HOME/myexpos/xfs-interface
ubuntu@arunpravinap24710:~/myexpos/xfs-interface$ ./xfs-interface
Unix-XFS Interface Version 2.0.
Type "help" for getting a list of commands.
# load --int=timer $HOME/spl/spl_progs/sample_timer.xsm
Can't open source file.
File r ❖❖ not found.
Error while trying to delete temporary file
# load --int=timer $HOME/s^Z
[1]+  Stopped                  ./xfs-interface
ubuntu@arunpravinap24710:~/myexpos/xfs-interface$ ./xfs-interface
Unix-XFS Interface Version 2.0.
Type "help" for getting a list of commands.
# load --int=timer $HOME/myexpos/spl/spl_progs/sample_timer.xsm
# exit
ubuntu@arunpravinap24710:~/myexpos/xfs-interface$ cd $HOME/myexpos/xsm
ubuntu@arunpravinap24710:~/myexpos/xsm$ ./xsm --timer 2
```